

E-Commerce US Dataset Analysis

Report 2: Data Integration

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1. Objective

- Combine 8 separate source tables into two analytical tables: master_orders and items_enriched.
- Preserve the 10,000-order grain — no row explosion at any stage.
- All joins use LEFT strategy to retain all orders, including canceled ones.

2. Pre-Merge Aggregations

Before joining, all one-to-many tables were collapsed to one row per order:

Aggregated Table	Source	Output Rows	Key Columns Created
items_agg	order_items	10,000	n_items, total_price, total_freight, avg_price, n_unique_products, n_unique_sellers
pay_agg	order_payments	10,000	total_payment, max_installments, n_payment_types, main_payment_type
rev_agg	order_reviews	9,332	avg_review_score, n_reviews
geo_agg	geolocation	849	geo_lat, geo_lng, geo_city, geo_state
cat_agg	order_items + products	10,000	main_category, main_brand, n_category

3. Master Orders Build — Join Sequence

- Step 1: Start with orders table (10,000 rows).
- Step 2: LEFT JOIN customers on customer_id — adds demographic and geographic columns.
- Step 3: LEFT JOIN items_agg on order_id — adds item count and price aggregates.
- Step 4: LEFT JOIN pay_agg on order_id — adds payment totals and method flags.
- Step 5: LEFT JOIN rev_agg on order_id — adds review score (668 nulls for canceled orders).
- Step 6: LEFT JOIN geo_agg on zip_code_prefix — adds coordinates and city/state.
- Step 7: LEFT JOIN cat_agg on order_id — adds primary category and brand.
- Result: master_orders.csv — 10,000 rows × 36 columns. No row duplication at any step.

4. Items Enriched Build

- Start with order_items (21,800 rows).
- LEFT JOIN products on product_id — adds name, category, brand, cost, price, dimensions.
- LEFT JOIN sellers on seller_id — adds company name, city, state.
- Result: items_enriched.csv — 21,800 rows × 24 columns.

5. Output File Summary

File	Rows	Columns	Location
master_orders.csv	10,000	36	data/raw_integrated/
items_enriched.csv	21,800	24	data/raw_integrated/
relationship_mapping.csv	8	4	data/raw_integrated/

6. Master Table Column Groups

6.1 Order Lifecycle (13 columns)

- order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_estimated_delivery_date, order_delivered_carrier_date, order_delivered_customer_date.
- Plus 5 computed delivery metrics added during cleaning.

6.2 Customer Profile (8 columns)

- customer_unique_id, customer_name, customer_gender, customer_age, customer_zip_code_prefix, customer_city, customer_state, customer_segment.

6.3 Aggregated Financials (8 columns)

- n_items, total_price, total_freight, avg_price, total_payment, max_installments, n_payment_types, main_payment_type.

6.4 Review Metrics (3 columns)

- avg_review_score, n_reviews, has_review.

6.5 Location Attributes (5 columns)

- zip_code_prefix, geo_lat, geo_lng, geo_city, geo_state.

6.6 Product Attributes (3 columns)

- main_category, main_brand, n_category.

7. Relationship Mapping Document

- relationship_mapping.csv documents all 8 join paths used in the project.
- Each record captures: from_table, to_table, join_key, cardinality, and join_type.
- Serves as the authoritative reference for anyone reproducing the integration pipeline.

8. Key Outcomes

- Zero row explosion — order count held at exactly 10,000 through all 6 joins.
- 668 null review rows retained (not dropped) to preserve canceled order records.
- Single flat table (master_orders) enables order-level analysis without further joins.
- items_enriched retains row-level granularity for item-level analysis and pricing.